

SandSnap: Measuring and mapping beach grain size using crowd-sourced smartphone images

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ABSTRACT

Sediment grain size is a critical parameter for sediment mobilization and transport, but often has the highest uncertainty of any coastal sediment transport model input parameter. SandSnap is an initiative to engage the public to amass a beach grain size database by taking photos of the beach sand with a coin in the image for scale and uploading the image to a web application. Images are analyzed with two deep learning convolutional neural networks one to detect the coin and the second to measure the grain size, which is trained on sediment samples within the sand regime. The results for nine gradation metrics are returned to the user within 2 min of image upload. Results from 263 test images have a mean percent error of -6.5% and median absolute error of 22.4% for the median grain size (d_{50}) with a small fine bias of -0.042 mm. The use of the database is highlighted by applying SandSnap output as an input to the AeoLiS aeolian sediment transport model to predict coastal dune growth at a nearly national scale using the full eight grain size classes ($d_{10} - d_{90}$) from the SandSnap database. These outputs are used to inform the potential value of having spatially comprehensive grain size distribution information as part of coastal engineering design and planning. Education and outreach techniques for the SandSnap initiative are described in the manuscript. Though some challenges remain, the spatially and temporally robust beach grain size database being developed by SandSnap will help to improve numerous coastal engineering analyses including coastal resilience and vulnerability quantification, beach nourishment life cycle and uncertainty analysis, beach compatibility for the beneficial use of dredged sediment, and large-scale coastal morphology modeling.

1. Introduction

The coastal zone is a fragile ecosystem that is vulnerable to pressures from development and climate change. Nearly 40% of the United States (U.S.) population lives in a coastal county (NOAA, 2024a). High-density development, pollution, sea level rise, and more frequent storms are all potential threats to this region. Scientists, engineers, and community leaders depend on high-quality data to make effective management decisions to protect the coastal zone. In the U.S., there are nationwide depositories for coastal elevation (Eisemann et al., 2019), wave

characteristics (Bryant et al., 2016), water level, and tidal data (NOAA, 2024b). Although previous studies have compiled sporadic international beach grain size data (e.g., Bujan et al., 2019; Holland and Elmore, 2008; McFall 2019), there is no spatially and temporally “robust” – with both high frequency/density of entries and spanning large extents - beach grain size database in the U.S. or globally that uses standard methodologies and is regularly updated with new information. As a result, practitioners often must collect this information manually in the field for specific coastal hazards and engineering applications, or rely on sparse information from the literature to inform local site conditions.

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The lack of grain size data limits large-scale (100's of km) coastal morphological change studies, large-scale coastal resilience quantification, and opportunities to beneficially use dredged sediment. Large-scale shoreline change studies typically categorize the sediment as sand/gravel or non-sandy based on limited ground-truth data (Luijendijk et al., 2018; Vitousek et al., 2023; Vos et al., 2023). Coastal resilience metrics do not currently include grain size explicitly (Dong et al., 2018; Durkin and Chambers, 2019; Janssen and Miller, 2022; Spurgeon et al., 2023), even though grain size is a critical characteristic for sediment mobilization and transport (Shields, 1936; Soulsby, 1997; among others). Additionally, many coastal states require sediment placed on the beach to match the existing sediment gradation (American Shore and Beach Preservation Association & Coastal States Organization, 2023), highlighting the widespread need for up-to-date beach grain size properties. Without a robust grain size database to capture the full temporal and spatial sediment gradation variation, sediment dredged from navigation channels may not be allowed to be placed on the beach profile (McFall et al., 2021). Beyond the lack of a beach grain size database, many beach sampling plans for projects lack adequate samples to statistically capture the spatial and temporal beach gradation (Krumbein, 1953; Prodger et al., 2016; Turki et al., 2013). This increases the uncertainty for the grain size used in the practical application of numerical models (Soulsby, 1997).

Whereas large scale beach topography data can be collected using far-field remote sensing tools allowing the spatial scalability of measurements, current methods of detailed grain size information require the local collection of it. Therefore, a coordinated, national scale measurement of spatio-temporal beach grain size with traditional sample collection and sieve analysis would be both logistically complicated and costly. Crowd-sourcing data collection via citizen science is growing in popularity and can be a cost-effective research technique to amass more data than would be otherwise logistically feasible (Gura, 2013). Citizen science efforts have yielded broadly valuable data for topics such as the presence of invasive species (e.g., Encarnação et al., 2021), health and biosciences (e.g., Wiggins and Wilbanks, 2019), and across a broad range of earth science applications (Lee et al., 2020). Citizen science has also been used specifically for the coastal sciences, including for both acquiring data and educating the public about coastal processes (Harley and Kinsela, 2022). Many citizen science projects have successfully engaged the public to support coastal process research by taking beach profiles (Eberhardt et al., 2022), photographing the beach for shoreline extraction (Harley et al., 2019), and even using unoccupied aerial vehicles (UAV's) for shoreline and volumetric mapping (Ierodiaconou et al., 2022).

SandSnap is an initiative to create a spatially and temporally robust beach grain size database by engaging citizen scientists to collect sediment imagery using their smartphones, as shown in Fig. 1. SandSnap relies on the analysis of citizen-scientist-uploaded photos to estimate the grain size (McFall et al., 2023). Determining sediment characteristics from imagery has been attempted in several previous studies using statistical techniques applied to the image's pixel intensity such as auto-correlation (Buscombe et al., 2010; Rubin, 2004; Warrick et al., 2009), fractals (Buscombe and Masselink, 2009), semivariance (Carbonneau et al., 2004, 2005), and wavelet analysis (Buscombe, 2013). More recently machine learning techniques have been applied to determine grain characteristics (Buscombe, 2020; Lang et al., 2021). Collectively, the widespread usage of imagery for grain size analysis has been shown to result in grain size information comparable to sieving and other traditional forms of analysis and has led to spatially and temporally extensive grain size measurements that have shed light on sediment transport processes such as suspension threshold and settling velocity (e.g., Buscombe et al., 2014; Cuttler et al., 2017; Prodger et al., 2016; Rubin et al., 2020). To apply these techniques many previous studies required high-resolution cameras or specific equipment (Barnard et al., 2007; Gallagher et al., 2011; Lendzioch et al., 2023; Matsumoto and Young, 2023; Warrick et al., 2009). The resolution of cameras built into



Fig. 1. Example of a user uploaded SandSnap image.

smartphones has dramatically improved in recent years, which now allows for community members to collect sufficiently high-resolution imagery for sediment analysis (McFall et al., 2020).

This manuscript details the machine learning technologies that have been leveraged and packaged in a web-accessible form to enable the public to collect their own grain size information via SandSnap and that has enabled the development of a growing global database of robust beach grain size information. Section 2 describes the methods applied by the SandSnap web application to measure nine gradation metrics from beach sand imagery collected by smartphones and to record the location shared from the phone's Global Navigation Satellite System (GNSS), allowing for geotagged sand samples. This section also provides an example use of the data for aeolian transport modeling. Next, results are presented in Section 3 which assess SandSnap-derived grain size statistics relative to traditional grain size measurement approaches, presents the globally distributed measurements currently within the SandSnap database, and shows the results of a proof of concept set of simulations for demonstrating the value of a public database for coastal engineering modeling applications. Discussion of the results, outreach efforts to engage potential citizen scientists, and next steps are provided in Section 4. Conclusions are given in Section 5.

2. Methods

2.1. Web application and user interface

SandSnap (<https://sandsnap-erdcchl.hub.arcgis.com/>) is a web-based application for collecting and analyzing images to determine sediment grain size. The user can visit any coastal location, take a photo of the sand with a U.S. coin in the image, and upload the image for analysis. The uploaded image is analyzed within 2 min and the gradation results are returned to the user with a fun fact about a famous beach with a similar grain size. The uploaded image and calculated results for nine gradation metrics (d_{10} , d_{16} , d_{25} , d_{50} , d_{65} , d_{75} , d_{84} , d_{90} , and mean) are added to the SandSnap database and can be viewed on the data viewer.

Details of the image processing are described in the two subsequent sections, but the overarching workflow is noted here. After the user uploads the photo with coin to the application, it is processed with several pre-processing checks and filters. First a model determines if it is a SandSnap image (sand with a coin). Then a coin detection model scans the image to determine the size and location of the coin. The image resolution (mm/pixel) is determined using the estimated coin size in pixels and the known diameter of the coin. The image is also checked for blurriness by measuring the focus of the image. Each of these checks have thresholds for failure and the web application will return any failed checks to the web application dashboard. These initial image processing and coin detection steps that serve as pre-processing steps in the workflow (Fig. 2) are further described in Section 2.2. The portion of the image without the coin is segmented into 1024 x 1024 pixel subwindows and the most in focus subwindow is processed by the SediNet machine learning model (Buscombe, 2020) to measure the grain size, as is further described in Section 2.3. Results from the SediNet model are converted from pixels using the previously determined image resolution, then returned to the web application dashboard and saved on the database.

The user interface of the website was designed to be simple and intuitive. The website's landing page allows the user four options: "SandSnap Now!", "Learn More", "Explore Database", and "Get the Data". "SandSnap Now!" takes the user to the SandSnap submission form. The form directs the user to mark the location of the sample using their phone's GNSS, upload the image, indicate the coin type used, select the cross-shore morphological position on the beach, and add any notes like if the sample was taken before or after a storm. Because the submissions are intended to be submitted by non-technical members of the public, all instructions were written using very limited technical language. For the cross-shore morphological position, descriptions of the beach dune, berm, swash, and "other" are given in easily understood language (e.g., the berm is the dry part of the beach where you would set up your beach chair). Note that while SandSnap has been trained and tested on images from the subaerial beach, the app does not prohibit users from uploading imagery from the swash zone. The cross-shore morphological position is collected because the grain size change in the cross-shore direction can be dramatic (>300%) on sand beaches (Bascom, 1951; Gunaratna et al., 2019; Moutzouris, 1988), and it may also help with quality control.

The "Learn More" section of the website describes the project motivation, details about the database, instructions for collecting samples, a fun quiz about sand, and school engagement opportunities in the form of a science fair project description and a class lesson plan. The "Explore Database" section allows the user to see all the processed images, their location on a map, and their gradation results. Images and gradation results in the table automatically filter based on the viewing area of the map. Additional high-resolution imagery assembled by Professor Rob Holman, emeritus faculty of the Coastal Imaging Lab at Oregon State University, are shown on the map and can be downloaded. The "Get the Data" section of the website provides links and information about how the images and gradation information data can be downloaded for user defined areas of interest or for the entire globe.

2.2. Coin detection and image filtering

The next two sections detail the processing performed on each citizen-scientist uploaded image by the SandSnap web application. User uploaded images are saved to a directory that is swept for batch processing every 60 s by the SandSnap processing code in the US Army Corps of Engineers' (USACE) Microsoft Azure cloud computing environment. First each image is processed using a coin detection/image filtering routine. Occasionally SandSnap users submit images that do not align with SandSnaps purpose either by accident or on purpose. For example, a user might accidentally upload a screenshot or non-beach related photo from their camera roll. To ensure these images do not continue to be processed, the images are first checked with the binary classification MobileNetV2 (Sandler et al., 2018) model to verify the images have sand and an easily identifiable coin. The model retains the original architecture and weights, and is appended at the end with additional training layers. This SandSnap verification model was trained with 500 curated images that included bad images with common mistakes such as phone screenshots, beach settings, and ocean settings.

After passing the SandSnap verification model, a machine learning coin detection model is applied to identify the "coin" pixels – pixels that are determined to be displaying part of the coin rather than sand. The coin detection model is a neural network with multiple hidden layers, specifically a residual U-Net implemented using the Segmentation Gym software (Buscombe and Goldstein, 2022). A residual U-Net is a "fully convolutional" neural network U-Net architecture (Ronneberger et al., 2015), modified with residual or 'skip' connections (Drozdal et al., 2016). In a network architecture with residual connections, each layer gets to see both the output from the previous layer (standard) as well as the inputs to that layer. This increases model accuracy and training speed (Buscombe and Goldstein, 2022). SandSnap's coin detection model was trained with 179 SandSnap images containing 161 U.S. quarters, 8 U.S. dimes, 6 U.S. nickels, and 5 U.S. pennies. This ratio of coin types used in training is similar to the ratio of coin types used in image submissions on the database.

The focus of the original image is checked at this stage using the variance of the Laplace filter of the greyscale image (Pech-Pacheco et al., 2000). The discrete Laplace operator is calculated using a windowed approach on a Gaussian-filtered image as the sum of differences over the nearest neighbors of the central pixel, i.e., the spatial second derivative of the Gaussian-smoothed image. The use of the Laplace filter assumes that high variance is indicative of images with more edges, and therefore in greater focus. Images with low variance, less than 40, are discarded, and the submission is flagged as being out of focus.

If the detected coin diameter is less than 100 pixels, then the image is rejected with a message indicating that a coin could not be detected in the image. This threshold is enforced because detected small coins indicate the imagery has been collected too far away from the sediment surface, such that individual grains are indistinct. The coin detection process can be imperfect, so the cluster(s) of detected coin pixels are further analyzed. In an ideal SandSnap run, all predicted coin pixels form one contiguous circular or slightly ellipsoidal pixel cluster like the shape of a coin. However, this is not always the case. The coin detection

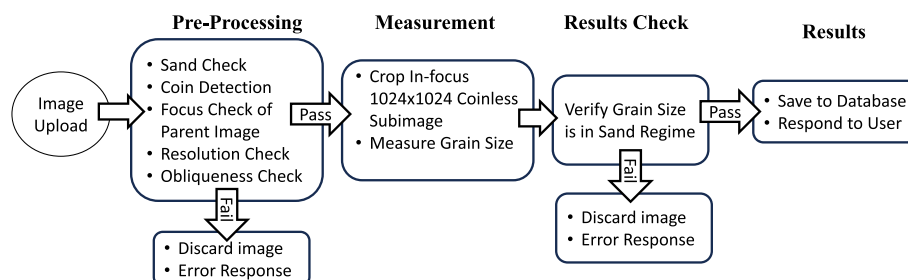


Fig. 2. Image processing workflow executed by the SandSnap web application.

model can return results where there are discrete coin pixel clusters isolated from one another in the image. This could be due to shell hash, rocks, debris, bad lighting, or image blurriness. Having multiple discrete predicted pixel clusters leads to inflated predictions of the coin diameter and therefore an inaccurate calculation of the mm to pixel ratio. A Connected Component Labeling (CCL) method with an eight-way connectivity input (two pixels sharing a common corner or edge) is used to identify and differentiate these discrete connected pixel clusters (Grana et al., 2009; He et al., 2017; Pearson et al., 2020; Song et al., 2017). The small discrete coin pixel clusters (<25,000 pixels by area) are removed. The remaining point cluster(s) (>25,000 pixels by area) are checked for uniformity by calculating the centroidal moment of inertia about the major and minor axis of the cluster. An ellipse fit is applied to each cluster to obtain the centroidal coordinates (x_0, y_0) and rotation angle (θ) of each connected component. A rotation matrix is applied as

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix} \quad (1)$$

where x and y are the pixel coordinates, θ is the rotation angle in radians, x_0 and y_0 are the centroidal coordinates, and x' and y' are the new rotated points. The centroidal moment of inertia about the x -axis is calculated as $I_x = \int y^2 dA$ and about the y -axis as $I_y = \int x^2 dA$. These equations can be further simplified since $dA = 1$ (the area of 1 pixel is 1).

The ratio between the major and minor moment of inertia provides a quantification of uniformity. Values closer to 1 are more circular and values farther from 1 are more oblique. Clusters with ratios larger than 1.25 are removed and the most uniform cluster is assumed to be the coin. If all the clusters have ratios greater than 1.25, the image is rejected because the image was taken too oblique to the sand surface or the coin detection model returned inaccurate results. Out of a dataset of 263 test images only a single image returned inaccurate coin predictions using a combination of the connected components function and the moment of inertia ratio. The diameter of the coin cluster in pixels is combined with known user-input U.S. coin diameter in mm to determine the resolution of the image in mm/pixel. Using these methods, the Mean Average Percent Error (MAPE) of the mm/pixel ratio was 3.60 for the 263 labeled test images. The model is capable of detecting non-U.S. coins, but the web application currently requires U.S. coins. There are plans to expand to other coins in the future. Non-U.S. users of SandSnap are presently encouraged to use a U.S. coin where possible.

If the image passes the above quality control checks then the coin portion of the image is cropped out such that only the largest rectangular “sand” portion of the image remains. When cropping out the coin portion of the image the detected coin diameter is scaled up by a factor of 1.05 to ensure no portion of the coin is retained in the remaining sand portion of the image. The size of the remaining sand portion of the image is checked to ensure it is larger than 1024 x 1024 pixels - the size required for SandSnap’s version of the SediNet grain size model. The processing routine then searches for the most in-focus 1024 x 1024 square window within the remaining sand portion of the images by computing the “edge score” of each sliding window across the image. The “edge score” calculation is based on the Canny edge detection algorithm (Canny, 1986). This approach operates under a similar assumption to the variance of LaPlacian approach, that a higher number of edges corresponds to greater focus. The 1024 x 1024 sand-only subimage with the highest edge-score is passed to SediNet for grain size calculation. The image focus is checked again in this stage, this time on the subimage. Subimages with edge scores less than 4800 are discarded and the submission is flagged as being out-of-focus. The edge score threshold of 4800 was selected because a significant increase in the grain size MAPE results was observed for subimages with edge-scores below that threshold.

A major focus of SandSnap is to avoid frustrating potential users with burdensome image quality requirements, under the assumption that citizen scientists are likely to stop using SandSnap if it is too onerous.

The checks for in-focus images are therefore deliberately lenient. All results but the most egregious, or those with deficiencies that preclude them from further processing, are returned to the user, including some results from blurry images. However, results from those blurry images are not saved in the grain size database. Furthermore, until further analysis is done on progressively out-of-focus images in a controlled setting, it is unclear the precise effect these have on our reported results apart from being highly detrimental below the selected threshold values. Currently, obviously blurry images are removed from the database by the site administrator, but results may be viewed in the “Recent Submissions” section of the web application. In a similar vein, a more sophisticated means to detect in-focus images (the Brisque score - Mittal et al., 2012) was tested in the coin detection/image filtering but was ultimately discarded in favor of the selected approaches due to processing speed considerations. Another major focus for SandSnap is to return results to users as rapidly as possible. The Brisque score filter required at least one order of magnitude greater processing time.

2.3. Digital grain size measurement

SediNet (Buscombe, 2020) uses machine learning to estimate grain characteristics. Models are trained using example images and associated grain size percentiles computed from sieve analysis. In addition to the grain size percentiles, the mean grain size, Φ_{mean} , is calculated as

$$\Phi_{mean} = \frac{\Phi_{16} + \Phi_{50} + \Phi_{84}}{3} \quad (2)$$

where Φ_{16} , Φ_{50} , and Φ_{84} are the 16th, 50th, and 84th grain size percentiles, respectively, in phi units (Folk and Ward, 1957). The mean grain size is then converted from phi units to mm. More than 35 organizations, consisting predominately of universities conducting coastal field measurements for other studies, were involved in the collection of imagery and physical samples to train the SediNet model for the SandSnap project. Many of these organizations will continue to collect and send additional imagery and physical samples for future model training. The model uses a deep learning neural network framework to learn the mapping between the image and available control grain size data measured via sieve analysis, and extracts only the image features that covary strongly with grain size. The SediNet model used for SandSnap was trained using 158 images collected from the 45 sites shown in Fig. 3 (McFall et al., 2022) with a range of mineralogy (e.g., quartz, carbonate). Each image was taken from a different altitude (thus has a unique scaling) and had an associated grain size distribution from a sieve analysis on a physical sample collected at the same time/location. To train the model, the coin scaling was performed manually, then the coin was cropped out of the 158 training images. The remaining portion of each image was sub-windowed into multiple 1024 x 1024 pixel training subimages, with 512 pixels of allowable overlap. The subimages were manually filtered prior to training. The total number of subimages used per training image depends on the resolution of the original parent image, the size of the coin within the image, and the number discarded during manual filtering. This process resulted in a total of 2458 1024 x 1024 pixel subimages to train SandSnap’s SediNet model, each with known scale in mm/pixel.

For the SandSnap web application, the trained SediNet model is run on the 1024 x 1024 pixel subimage output by the coin detection scheme described above. The d_{10} , d_{16} , d_{25} , d_{50} , d_{65} , d_{75} , d_{84} , d_{90} , and d_{mean} are determined in pixels for each user submitted image. The mm per pixel results from the coin detection step are used to convert the SediNet results from pixels to mm. The final results in mm are returned to the front-end dashboard for SandSnap users and logged in the grain size database.

2.4. Location measurement

SandSnap marks the location of the samples taken using the shared



Fig. 3. Locations where training imagery and physical samples were collected. Two additional sites that are not shown are from the Black Sea and Bondi Beach, Australia.

location from the smartphone's built-in GNSS location capability. The positioning accuracy of the GNSS module on a smartphone is generally between 3 and 5 m under open sky conditions and over 10 m under harsh multipath environments such as near buildings, bridges, and trees (Schaefer and Woodyer, 2015; van Diggelen and Enge, 2015; Zangenehjad and Gao, 2021). The SandSnap web application also allows the user to manually drop a pin on the map where the photo was taken when uploading a sample after the time of acquisition. This feature is required because cellular service is needed to use the web application for uploading the image, and service may not always be available in the field. Additional error in location may be introduced by this feature due to imprecision in user-input pin location. At the current time, we do not have a way to determine these potential inaccuracies on the somewhat rare occasion that no cellular signal is available.

On the web application, the location pin is set to a default location in the center of the continental U.S. If the pin is not moved from that location, a location error is triggered. A location error can also be triggered if the location pin is > 500 m from the shoreline in the ocean. When a location error is triggered, the image metadata is checked for location information, and if available, the recorded location is moved to the location noted in the metadata. This is only used for location errors. Location metadata is a feature that has to be enabled in the user's phone, and the majority of SandSnap submission images do not have location

metadata.

To test the location accuracy that can be expected using the SandSnap web application, a small experiment was conducted in Duck, North Carolina, U.S. The goal of the experiment was to help quantify total location error (aleatory and epistemic) collecting location information from the public using the web application on the beach. A total of 25 volunteers were asked to take a SandSnap and mark their location in the web application at a surveyed point. Participants were from diverse locations around the U.S., used both Android and iPhone smartphones, and used three different national cell phone service providers.

2.5. Aeolian application

Grain size is a critical parameter for a broad range of coastal science and engineering applications related to topics such as beach erosion, beach nourishment design, coastal profile modeling, and coastal hazard quantification. Given that the SandSnap platform has been trained and tested predominantly on grain information on the subaerial beach, these data also well represent the source of sediment available for wind-blown transport in coastal environments. Coastal foredunes are prevalent landforms on much of the world's stable or prograding sandy coastlines that are predominantly formed due to wind-driven sediment inputs.

Due to their high topography which can limit flood damages during

high water events, such as hurricanes and Nor'easters, dunes are also widely constructed in managed coastal settings. The ability for constructed dunes to grow with time and not be lost to the ocean, is, however, highly dependent on local site parameters such as the local beach slope/width and environment characteristics. The beach grain size is similarly important given that it influences the threshold velocity for the initiation of wind blown sediment transport, local sediment availability, and the magnitude of landward-directed sediment fluxes into the dune complex under onshore-directed winds (e.g., Bagnold, 1937; Delgado-Fernandez, 2010). As a demonstration of the utility of a public grain size database for applied scientific and engineering applications at a regional scale, we show outputs linking the SandSnap database with the AeoliS (Hoonhout and Vries, 2016) model to provide nearly national scale predictions of potential aeolian sediment fluxes into coastal foredunes around the U.S. coastline.

AeoliS is an open-source, process-based numerical model that simulates multifraction aeolian transport rates for supply limited environments, such as for the coastal zone. Recent development of the model has focused on incorporating more complex moisture dynamics (Hallin et al., 2023), vegetation growth and expansion (Meijer, 2020), vegetation shear coupling capabilities (Dickey et al., 2023), topographically-induced flow acceleration (van Westen, 2018), dune erosion capabilities (Heminway et al., 2024), and improved aerodynamic roughness parameterizations. These numerical capabilities, as well as validation against data, continue to support its increasing use for engineering-type applications. Here we use SandSnap outputs as an input to AeoliS over a large geographic region to demonstrate both the local (e.g., aeolian sediment inputs) and national scale (regional variability in sediment fluxes) utility of robust grain size distribution measurements.

The SandSnap database was downloaded from the public ArcGIS database as a comma separated value (CSV) file. The database was subset to only include values within the continental U.S. (CONUS) along the ocean coastline, where consistent morphological data and environmental data are present. Data points with a "Location on Beach" descriptor of "other" were excluded in order to only utilize samples from swash, berm, and dune systems. For samples meeting these criteria, first the closest location where both beach slope and dune morphology data (dune toe elevation, dune crest elevation, dune slope) was available from a CONUS-wide database in Cohn (2022). A synthetic beach-dune profile with a linear beach slope was developed using these local morphometrics. The closest USACE Wave Information Studies (WIS) output node was used to derive offshore wave properties and wind forcings for each model. The closest tide gauge operated by the U.S. National Oceanic and Atmospheric Administration (NOAA) was sourced from the NOAA Tides and Currents database. From these environmental inputs wave height and wave period are used to estimate wave runoff following the approach of Stockdon et al. (2006) internally to AeoliS. Wave runoff is combined with the still water level time series, as recorded from the NOAA tide gauge, within AeoliS to define the total water level, which determines wet and dry cells in the model. Aeolian transport is calculated by AeoliS in each dry cell based on time variable wind and direction forcing, with the wind direction being modified into a shore normal angle based on the local shoreline azimuth for each site. Hourly environmental time series from WIS and NOAA were used to force the model for the period from 1 January 2018 to 31 December 2018. Model default parameters were used in this application except that vegetation is turned on in the model and assumed to be present at all grid nodes landward of the defined dune toe.

A total of 897 sites met the criteria for available grain size, morphology, and environmental data in the U.S. Multifraction simulations using the 8 grain size classes (d_{10} – d_{90} , omitting d_{mean}) from SandSnap were input as a full grain size distribution into AeoliS. Depending on the profile length for each specific site, each annual-scale model simulation took between about 5 and 10 min. The total annual volumetric change of the dune (ΔV_{dune}) was calculated from AeoliS bed

elevation outputs, where the volume change is integrated from all cells landward of the initial dune toe location, and is the primary metric reported in this work. These outputs are used to comment on the value of robust grain size availability and demonstrate the feasibility of numerical model scalability when consistent and reliable physical data is available across broad geographic stretches.

3. Results

3.1. Accuracy of granulometry measurements from smartphone-derived images

The overall accuracy of the SandSnap model results were tested on 263 test images from 113 different sites exhibiting a wide range of beach sedimentology, phone cameras, and users, all of which have corresponding physical sample sieve analysis results for validation. All the test images are from either the beach surface or within 5 cm of the beach surface. Many sites are identical to those used to collect training data for the models, however in those instances a separate input image from the same site is used to test the models' accuracy. Fig. 4 displays 1-1 plots of the sieve-measured (a) d_{10} , d_{16} , d_{25} , (b) d_{50} , d_{mean} , d_{65} , (c) d_{75} , d_{84} , d_{90} (x-axis) compared to the SandSnap measurements of the same quantities (y-axis) for the 263 input images that completed processing.

Table 1 contains overall error statistics for each grain size metric, specifically the bias in the grain size estimate, the mean percent error, and the median absolute percent error. The bias is calculated as $\frac{1}{n} \sum_{i=1}^n (d_{pi} - d_{ai})$, where d_p is the SandSnap-predicted grain size metric and d_a is the sieve-measured value. The mean percent error and median absolute percent errors are similarly calculated as $\frac{100}{n} \sum_{i=1}^n \frac{(d_{pi} - d_{ai})}{d_{ai}}$ and $med\left(\frac{abs(d_{pi} - d_{ai})}{d_{ai}}\right)$, respectively. Note that all reported errors are the

combined error of the machine learning-based grain-size estimation model and the image scaling procedure using a machine learning-based coin detection model. Due to the summation of these independent errors in quadrature, misspecification of both grain size percentile and pixel size can result in relatively large combined errors. It should also be noted that additional sources of error could exist, such as image distortion, which are discussed below.

Table 1 shows that SandSnap is biased low for all grain size metrics. This bias is relatively small but increases in magnitude as gradation percentiles increase (from -0.024 mm for d_{10} to -0.130 mm for d_{90}). This finding of small but consistently negative bias is confirmed visually in the 1-1 plots of Fig. 4 and by the observation of small but negative mean percent error for all metrics. A fine bias contrasts with many previous studies that measured grain size from imagery (Buscombe et al., 2010; Matsumoto and Young, 2023; Warrick et al., 2009). Training data for most grain size models is measured with sieves, which measure the b-axis of the sediment aggregate – where the a-, b-, and c-axes are the longest, intermediate, and shortest axes of an ellipsoid-shaped grain (Garefalakis et al., 2023). In contrast, image measurements use the visible cross-section. This means coarse bias is more likely for image-derived grain size analysis but this is not the case for the current version of SandSnap. The accuracy of the SandSnap will improve with the inclusion of larger training datasets into the model, likely reducing the fine bias present in the current version.

The mean percent error is also negative for all grain size metrics and the magnitudes are relatively small, 9% or less for all 9 measured grain size metrics (Table 1). This is consistent with the bias already discussed and the observed behavior in Fig. 4 – the errors are close to evenly distributed about the 1-1 line with a small fine bias. The more extreme outliers exist for small subset of samples with larger overall grain size, and in each case the predictions are substantially smaller than the actual grain size values – likely influencing the predictions towards a fine bias. This is likely due to the relative scarcity of training and testing data for sediment with d_{50} exceeding 1 mm for the current version of SandSnap.

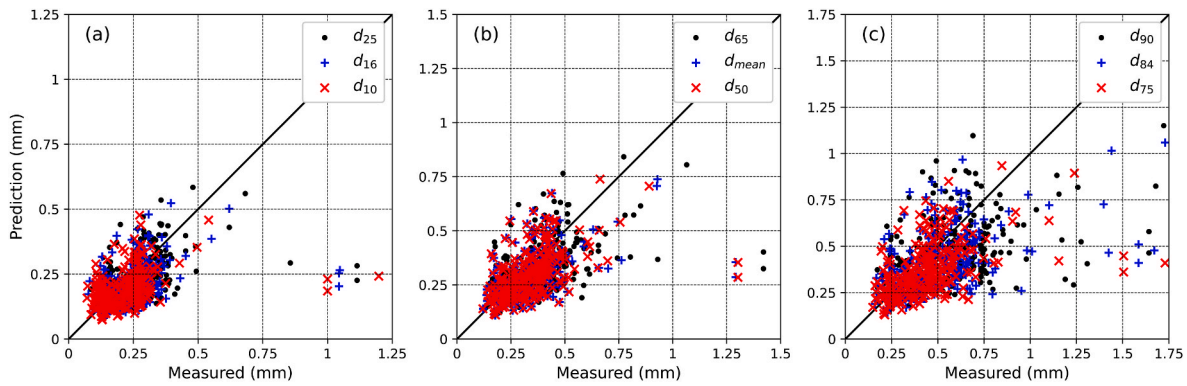


Fig. 4. Measured versus predicted results for all nine gradation percentiles determined using SandSnap.

Table 1

Bias and error quantification for nine gradation percentiles using 263 test images.

Metric	Bias (mm)	Mean Percent Error (%)	Median Absolute Percent Error (%)
d_{10}	-0.024	-0.1	26.1
d_{16}	-0.027	-1.8	25.4
d_{25}	-0.029	-3.4	24.3
d_{50}	-0.042	-6.5	22.4
d_{mean}	-0.046	-7.2	22.9
d_{65}	-0.055	-7.6	23.2
d_{75}	-0.073	-8.6	25.6
d_{84}	-0.101	-8.7	26.5
d_{90}	-0.130	-7.8	29.2

The mean percent error increases in magnitude from d_{10} to d_{84} but decreases slightly for d_{90} . This is visually supported by the behavior in Fig. 4. Fig. 4(c) demonstrates that outliers where the predictions are substantially smaller than the true grain size are more prevalent for larger grain size metrics (d_{75} , d_{84} , d_{90}) than the smaller metrics presented in Fig. 4(a) and (b).

The median absolute percent error ranges from 29.2% for d_{90} to 22.4% for d_{50} (Table 1). It is generally larger for metrics at the tails of the grain size distribution (d_{10} , d_{16} , d_{75} , d_{84} , d_{90}), reflective of generally higher uncertainty at the tails of the distribution, though it is largest for the largest grain size metrics. This behavior is distinct from that observed in the bias and mean percent error, which tended to increase with increasing grain size metrics. The median absolute percent error is only concerned with the magnitude of the errors rather than the sign and is higher than the magnitude of the mean percent error for all grain size metrics. The median percent error will likewise likely be reduced with the inclusion of additional training data for sediment with d_{50} exceeding 1 mm.

3.2. Location accuracy

The location accuracy using the SandSnap web application on a beach was tested by 25 participants. Of the 25 submissions in the location experiment, one triggered an error location by appearing in the middle of the Atlantic Ocean. The remaining 24 submissions had a mean absolute location error of 12.7 m. The submitted locations are shown in Fig. 5. Note that the background satellite imagery does not represent the beach state at the time of data collection – all images were collected in the subaerial portion of the beach. The total location error measured in this experiment using the web application on the beach accounts for both epistemic error affiliated with varying phone types, hardware, and location processing, and the aleatory error added by the users. Although the experiment had limited submissions, the results are qualitatively similar to the location accuracy experienced by the authors collecting



Fig. 5. Location results using the SandSnap web application (o) relative to a surveyed position (□) on the beach.

samples.

All 25 submissions in the location experiment noted the correct cross-shore morphological position, the beach berm. Considering the extensive cross-shore sorting of the sediment on the subaerial sandy beach (Bascom, 1951; Gunaratna et al., 2019; Krumbein, 1953; Moutzouris, 1988), the successful categorization for the sample's cross-shore morphological position is very useful. Longshore grain size variability is also important on sand beaches (Gallagher et al., 2016) and is easily observed on mixed beaches with gravel, cobbles, and shell hash (Buscombe and Masselink, 2006; Holland and Elmore, 2008; Matsumoto and Young, 2023; Nolan et al., 1999). SandSnap is currently trained exclusively with sand. Also, considering the rapid development in smart-phones over the last decade, it is reasonable to assume the phone's location accuracy will improve in the future.

3.3. Database

Creating a spatially and temporally robust database of beach sediment gradation is the main goal of the project. SandSnap initially consisted of a website where users uploaded an image and recorded their location. The interactive public web application was released in March

2022 that returned results to the dashboard. The release of an interactive web application coupled with an increase in publicity and outreach initiatives created an exponential increase in the user submissions to an average of 114 submissions per month in 2023. By 24 March 2024, the database had 2306 successfully processed submissions from 38 countries, 80% of which were from the U.S., as shown in Fig. 6. The web application currently requires a U.S. coin and most of the outreach was U.S.-focused, so the predominance of U.S. submissions is expected. As previously noted, all of the beach gradation, location, and image data can be viewed and downloaded from the database.

3.4. Aeolian application

Beach grain size is one of the dominant controls on sediment mobility by wind. Simulations of wind-driven sediment fluxes using the AeoliS model utilizing available grain size, morphological, and environmental inputs from 897 sites, demonstrate that there is broad variability in anticipated dune growth rates across the coastal U.S. (Fig. 7). Using standard model default settings for aeolian transport coefficients and the full grain size distribution at each site, median wind-driven ΔV_{dune} of $6.7 \text{ m}^3/\text{m}/\text{yr}$ were simulated, neglecting potential sediment losses due to wave-driven collision. The largest rates of dune growth were generally simulated along the U.S. Pacific Northwest coastline which are characterized by low gradient beach slopes, fine grained sands, and an energetic wind climate (Fig. 7; e.g., Ruggiero et al., 2016). Conversely, there is broad local variability in potential wind-driven growth rates along the U.S. Gulf of Mexico and Atlantic Ocean shorelines associated with narrower beach widths, broad grain size distributions, and event driven (less chronic) wind forcings above the threshold velocity for sediment mobilization. For example, simulated ΔV_{dune} vary from less than $1 \text{ m}^3/\text{m}/\text{yr}$ to over $5 \text{ m}^3/\text{m}/\text{yr}$ within the state of Florida.

4. Discussion

4.1. Variables affecting accuracy

Measurement accuracy is critical for the success of this project, and there are several variables that affect the accuracy. The image resolution varies between submitted images based on the distance from the ground and the camera resolution. Fig. 8 displays the image resolution as (a) mm per pixel (x-axis) and (b) the pixels per d_{50} sediment grain size (x-axis) plotted against the absolute percent error (y-axis) for the d_{50} of the 263 verification images. It describes the relationship between the resolution of the user-input image and the error from SandSnap, both in terms of the physical size of area covered in the image (mm per pixel) as well as relative to the physical size of the sediment grains displayed in

the image (pixels per d_{50}). Only the d_{50} is displayed here but the results for the other 8 metrics are similar.

Fig. 8(a) shows a trend between the image resolution and the percent error in the d_{50} measurements. For high resolution images (corresponding to smaller mm/px values) the percent error tends to be negative and becomes less negative as the image resolution decreases (larger mm/px values). At mm/px values around 0.03 this negative bias becomes nearly zero, and the results tend to be biased high for images with mm/px values > 0.03 . It is interesting that the SandSnap results tend to be biased low for extremely high-resolution images ($< 0.03 \text{ mm/px}$). This may be due to intra-grain texture (i.e., the variation from grain facet to grain facet) which at high resolution can bias size statistics for finer grains low (based on numerical simulations performed by Buscombe and Rubin, 2012). Similarly, the grain size tendency to be biased high for images with lower resolution has previously been reported by Warrick et al. (2009) and Buscombe et al. (2010) who attributed this to the tendency of adjacent grains to visually coalesce due to less distinct boundaries.

Fig. 8(a) also suggests that to achieve the best results the resolution for SandSnap imagery should be on the order of 0.03–0.04 mm/px, which corresponds to an image viewing area of approximately 7500 mm^2 , or slightly larger than a U.S. business card. Viewing areas smaller than this tend to be more out of focus and may suffer from sample size limitations, that is the number of individual grains in the image is insufficient for an accurate grain size measurement compared to a sieving procedure that samples more grains. For areas larger than the suggested optimal size, the overall worsening resolution starts to negatively affect grain size results. At the largest of scales, the image may be sampling a slightly different population of grains than were sampled and subsequently sieved. Additionally, at large scales, the effects of camera lens distortion start to have a greater effect due to collecting imagery at greater distance from the camera, making the estimated pixel scale unrepresentative of larger portions of the image. Fig. 8(b) therefore unsurprisingly indicates that the SandSnap results with the largest positive (and overall) percent error occur for images where very few pixels span the d_{50} of the sediment in the image. The percent error decreases until the px/grain (d_{50}) ratio reaches approximately a value of 9 at which point the presence of large positive outliers (images with percent error $> 80\%$ in magnitude) are eliminated. Images with px/grain (d_{50}) ratio greater than 9 display a negative bias, and the percent error becomes increasingly more negative until it levels off at -80% .

The SandSnap models are trained exclusively with sediment within the sand regime (median grain size from 0.063 mm to 2 mm; Wentworth, 1922). It is therefore not appropriate to measure the gradation of mixed beaches with sand and gravel for the simple logistical issue that in order to have high enough resolution to measure sand accurately, the

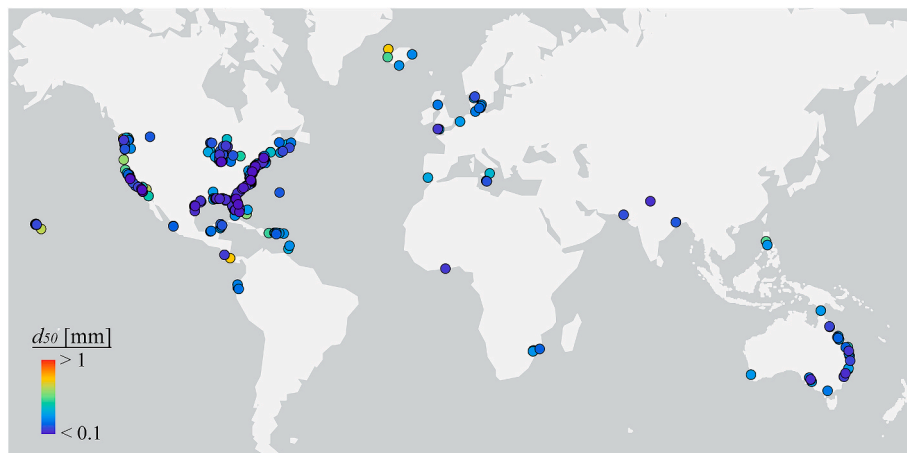


Fig. 6. Locations of SandSnap submissions in the global database as of 24 March 2024.

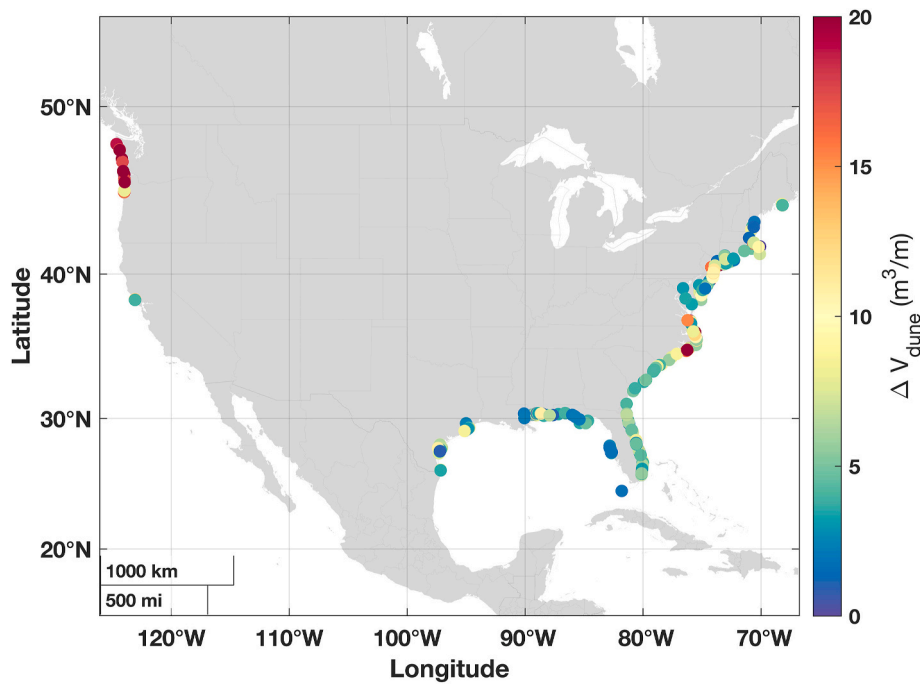


Fig. 7. Simulated annual ΔV_{dune} from AeoliS along the coastal U.S.

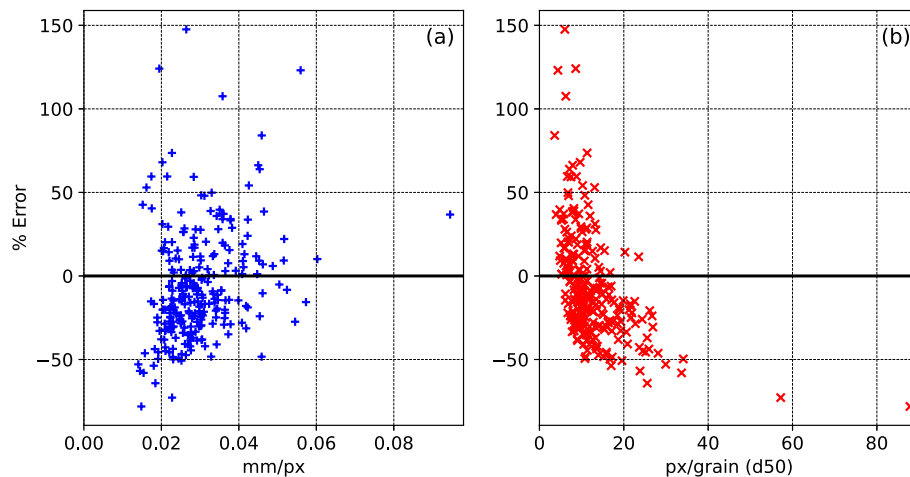


Fig. 8. Percent error relative to the image resolution in (a) mm/pixel and (b) pixel/ d_{50} .

viewing area of the image is not large enough to capture a representative sample of gravel. Preliminary tests with a few gravel aggregates in images tended to trigger quality assurance flags in the coin detection procedure by identifying multiple “coins”. Users are therefore advised to only submit images of sand, as the name ‘SandSnap’ specifically suggests.

The accuracy of SandSnap results and image focus using the “edge score” (not pictured) was investigated. SandSnap results have high errors for images with extremely low edge scores (<4800), but there is no obvious relationship between the error and image focus above this very low threshold. In our testing, extremely low edge score images are only achievable in modern smartphone cameras with deliberate attempts to take very out-of-focus images. As noted above, camera lenses suffer from distortion and the effects of this distortion cause greater errors in the specification of pixel scale at relatively high altitude. We make no attempt to correct imagery for lens distortion, so this remains an unquantified source of error.

The total error reported in Table 1 (i.e., image scale error and grain

size estimation error, summed in quadrature) is the most useful error metric because it notes the total potential error associated with the intended engineering and scientific uses of the estimated gradation values. It also includes both systematic (image processing methodology) and aleatoric errors (randomness from people, their phones, etc.). Random errors can be reduced by averaging (Spasiano et al., 2021) and systematic errors can be removed when identified (Lahoz and Schneider, 2014). It is important to note that the SediNet model error for grain size estimation is expected to decrease as training data continues to be expanded and smartphone imagery continues to improve.

4.2. Engaging the public

Successful engagement with the public is necessary for any citizen science program. Citizen science programs can help build resilient communities by fostering collaborations, increasing knowledge sharing, and building community trust (Conrad and Hilchey, 2011; Cigliano et al., 2015; Dean et al., 2018). Additionally, community involvement

can improve monitoring efforts by increasing the spatial and temporal extent of the monitoring and improving rapid response to episodic events. This final benefit is particularly effective when monitoring coastal change in response to storms.

Contributors to SandSnap appear to fall into two categories: coastal professionals and the public. Outreach initiatives for coastal professionals such as engineers, scientists, beach managers, and researchers have been at coastal-centric conferences [e.g., American Shore and Beach Preservation Association (ASBPA) National Conference, International Conference of Coastal Engineering, Coastal Sediments Conference, American Geophysical Unions-Ocean Sciences Meeting] and with coastal focused organizations [e.g., Coasts, Oceans, Ports, and Rivers Institute (COPRI); ASBPA]. Coastal professionals are motivated to collect SandSnap samples because it is an effective way to collect sediment data for their own projects. Several engineers have noted using SandSnap to quickly measure grain size information for preliminary or feasibility studies, and researchers have reported their interest in using grain size data to test hypotheses. It should be noted that machine learning-derived grain size measurements, including SandSnap, do not currently meet any universally accepted engineering standard (e.g., American Society of Civil Engineers, ASTM International). In the future, machine learning-derived approaches may have an engineering standard developed and the range of professional applications for the data could be expanded.

Outreach efforts to engage the public have been multifaceted. Collaborations have been successful with existing beach focused citizen science programs like NOAA's California SeaGrant programs CoastSB and Beach Watch. Adding SandSnap signage adjacent to CoastSnap (Harley and Kinsela, 2022) stations and installing flyers on bulletin boards near beach access points are currently being piloted.

The educational outreach for SandSnap includes developing library science kits, Girl Scout Citizen Science journey, science fair projects, and a class lesson plan. These educational initiatives will be extensively detailed in another outreach focused manuscript but are briefly described here for completeness. The library science kits consist of a large anchor bag that contains the instructions and materials to conduct four STEM activities on the beach and are designed for 5–9-year-old children. Coastal library patrons can check out a kit, complete the activities on the beach, and return the kit to the library for the next person. Three activities in the kit focus on studying the physical principles that govern coastal dynamics, including settling velocity, angle of response, grain size, and sediment composition. The fourth activity is to take a SandSnap to measure the grain size. A similar Girl Scout journey engaged the same age group. The Girl Scout activity focused on making observations about water and sand interaction on the beach, using the scientific method, and collecting a SandSnap sample.

The class lesson plan and science fair projects are linked on the SandSnap website. The class lesson plan uses the SandSnap database and is designed to address the Next Generation Science Standards (NGSS) for Earth Sciences in middle school. However, this activity could be used in high schools by varying the amount of student support provided by the instructor. The science fair project deploys students to the beach to take a SandSnap and measure the beach slope to study the grain size to beach slope relationship.

4.3. Possible future work

The SandSnap project relies on good grain size-estimation and good image scaling, both of which use image processing procedures that are reliant on data-driven models. As such, grain size estimates could improve if either or both procedures were improved. Given the pace of machine learning and its subfields, it is conceivable that in the future, new algorithms could analyze the image using a single model that scales the image and sizes the grains simultaneously. Even within the present project framework, we would expect continuing improvement with additional training data, for either of the deep-learning-based models

that the project relies upon. One of the strengths of our approach is that our models can in theory only get better with additional training data. Additionally, the existing framework is configured to allow additional models to be easily included such as models trained for specific geographic regions or specific sediment types.

To reduce the aleatory uncertainty affiliated with uploaded images, models are currently being evaluated for debris identification, shell hash classification, and coin categorization. Natural and anthropogenic debris (logs, vegetation, trash, etc.) are commonly found on beaches and have been included in user submitted images. A debris identification model would allow sandy areas to be identified for measurement and with a robust enough dataset, a debris model may even be able to qualitatively track debris types and location. Shell hash is often larger than the sand regime (>2 mm) and is not currently represented in the SandSnap model training data. A shell hash classification model could flag images and a another SediNet model that is extensively trained with shell hash could measure the gradation. SandSnap users currently input the coin used, and erroneously selecting the coin type will incorrectly change the calculated image resolution (mm/pixel). A coin categorization model could prevent this error in the future. Additionally, future model development could focus on other sedimentary characteristics known to influence sediment transport, such as grain shape and mineralogy.

It is not currently possible to quantify the number of SandSnap submitters because SandSnap submissions are anonymous. This makes it difficult to determine the efficacy of specific outreach efforts, but the exponential increase in SandSnap submissions is a positive indicator that the number of participants is increasing and outreach efforts have been successful. As the database submissions continues to increase, data sparse areas may be targeted by engaging local professional organizations and posting SandSnap signage at beach access points. Another potential improvement to expand the database is to allow non-U.S. coins in the processing workflow. This would significantly improve the feasibility of amassing a global beach grain database and make it much easier to collect a SandSnap on non-U.S. beaches. Additionally, further training, testing, and validation on swash zone imagery could allow SandSnap to further explore the effect of moisture content on the grainsize predictions.

5. Conclusions

Widespread beach sediment measurements are rare, owing in part to the labor associated with sample acquisition and processing. Thus, while wave, tide, and morphology information are generally available at a national scale in the United States, this gap of sediment size data has played into the inability to complete regional to national scale assessments of coastal sediment transport, erosion, and related hazards. There is a need for a spatially and temporally robust beach grain size database. The SandSnap initiative is successfully engaging citizen scientists to measure the beach sand grain size at broad spatial scales. This is made possible with the proliferation of smartphone ownership and continually improving smartphone cameras. Results from 263 test images show a mean percent error of -6.5% and median absolute error of 22.4% for the median grain size (d_{50}) with a small fine bias of -0.042 mm. The mean percent error is less than 9% for all nine measured grain size metrics. The higher uncertainty in the tails of the distribution cause increased absolute error in the gradation metrics farther from the mean.

The continually growing SandSnap created database can capture the spatial and temporal sediment grain size variation on sand beaches. It can be used to address numerous scientific questions including large-scale (100–1000's km) beach morphological studies, improve empirical depth of closure methods, improve beach nourishment life cycle analysis and uncertainty quantification, improve dune evolution modeling, and quantify the full range of beach sediment gradation to increase opportunities for the beneficial use of dredged sediment from navigation channels for beach nourishment. As a proof of concept, the

SandSnap database was used to show the ability to scale engineering models when widespread grain size data inputs become available, thus reducing the burden for setting up research tools. Crowd-sourcing the data collection engages the public in coastal science and cost effectively amasses a large amount of information. SandSnap can provide a tremendous benefit for the public and the scientific community by creating a robust database of quantitative grain size measurements to improve the understanding of the coastline's composition and seasonal variability. Furthermore, public outreach and educational opportunities utilizing SandSnap include engagement with existing coastal citizen science volunteers, library science kits, Girl Scout Citizen Science journey, science fair projects, and class lesson plans. SandSnap serves as a mechanism to engage citizen scientists to collect valuable beach grain size information for scientific and engineering workflows, and educate the public on coastal science.

CRedit authorship contribution statement

Brian C. McFall: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **David L. Young:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Shelley J. Whitmeyer:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Daniel Buscombe:** Writing – review & editing, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Nicholas Cohn:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Jacob B. Stasiewicz:** Writing – review & editing, Software, Methodology, Investigation, Formal analysis, Data curation. **Janelle E. Skaden:** Writing – review & editing, Software, Methodology, Formal analysis. **Brooke M. Walker:** Writing – review & editing, Methodology, Investigation. **Shannon N. Stever:** Writing – review & editing, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The web application is freely available and readers are encouraged to visit it at <https://sandsnap-erdcchl.hub.arcgis.com/>. The training data (images, subimages, and sieve measurements) are available at <https://zenodo.org/records/7063226>. The SediNet model code is available at <https://github.com/DigitalGrainSize/SediNet>.

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